

Name: Sudha Sree, Yerramsetty

Title of chapter: Chapter 04 - Data Sources and Biases

Summary:

This chapter looks at where data for machine learning originates from, the strengths and drawbacks of various sources, and the biases that jeopardize the validity of ML systems. Data can be gathered from purposefully collected sources (such as surveys or experiments), administrative records, social media, crowdsourcing, or data augmentation. Each source has distinct challenges: census surveys may have non-responses or manipulation, administrative data frequently offers proxies rather than genuine labels, social data can be noisy and biased toward certain populations, and crowdsourced labels may be inconsistent or contextually inaccurate. Even data augmentation and synthetic generation can result in distortions. The key insight is that most data utilized in machine learning is repurposed and was not initially created for the system's problem, therefore teams must carefully assess quality, relevance, and hidden biases.

The second half of the chapter provides an outline to think about bias and validity across the ML lifecycle. Biases are classified based on the type of validity they threaten: social bias (bias from human judgment, whether implicit or explicit) affects construct validity (whether or not the data measures what it should), representation bias (sample not representative of all populations) and temporal bias (dataset goes out of sync with reality due to shifts over time) harm external validity (whether the results generalize to other populations), while data preparation bias (bias introduced during cleaning, enrichment, or proxy selection of data) and data poisoning (bias resulting from malicious attacks on datasets, such as injection or manipulation) endanger internal validity (whether there are any errors in data processing). The authors emphasize that biases might result from measurement errors, insufficient/poor sampling, flawed preprocessing, or even deliberate malicious attacks. The chapter concludes that defining data sources and verifying for bias are critical processes; taking shortcuts here leads to unreliable, unfair, or even potentially hazardous ML systems. Diverse teams, robust appraisal, and defensive modeling tactics are advocated for bias reduction.

What are two points that you learned after reading this chapter?:

- The task definition stage explains exactly what the ML system is expected to do. One of the most prevalent causes of failure is a misdefinition of the task. The majority of data in ML systems is reused rather than being acquired expressly for the job, resulting in frequent and hazardous mismatches between the data and the true problem definition.
- No matter how careful one is, there is no such thing as a perfectly unbiased dataset. However, detecting and correcting biases prior to modeling improves results. Biases correspond directly to three categories of validity: construct, external, and internal. This mapping provides a systematic approach to understanding how data quality concerns affect machine learning systems.

What are two questions that you have about this chapter?:

- How can teams accurately detect subtle social biases in features or labels, especially when those biases are strongly established in cultural context? Even if someone manually tries to detect them, how can we make sure they are unbiased in all aspects? How do we even know for sure that not having any bias in every situation is the best option? There certainly are situations that actively benefit from some kind of "good" bias, aren't there?
- What practical safeguards can businesses put in place to protect against data poisoning attacks without excessively limiting data access or usability?

Do you have some initial thoughts or insights to approach the questions?:

- Detecting social bias might require the involvement of domain experts and diverse stakeholder panels, in addition to data scientists. Automated fairness indicators can help identify inequities, but qualitative insights from individuals who understand the social context are essential.
- To combat data poisoning, businesses should combine robust anomaly detection, secure pipelines, and differential data validation checks. A layered defense strategy (technical and procedural) could help strike the optimal compromise between usability and security.